Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

908
7 0.78%
Feb 20th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1019
7.00 0.69%
Feb 20th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

791
0 0.00%
Feb 20th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

126.42
0.00 0.00%
Feb 20th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

137.35
0.00 0.00%
Feb 20th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

990
-8 -0.80%
Week Ending Feb 17th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2305 (Jan) RMB/t (3pm close)

894.00
4.50 0.51%
Feb 20th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
March 23 USD/dmt

127.75
2.45 1.96%
Feb 20th, 2023 (5.30 pm Print)

SHFE Rebar
RB2305 (Jan) RMB/t

4184
17 0.41%
Feb 20th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

16.11
-0.06 -0.38%
Feb 17th, 2023

C5, W. Australia - Qingdao USD/t

6.06
-0.01 -0.25%
Feb 17th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4180
40 0.97%
Week Ending Feb 17th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

137.11
1.02 0.75%
Week Ending Feb 17th, 2023

Steel Inventory in China
million tonnes

19.88
0.46 2.36%
Week Ending Feb 17th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4240
80 1.92%
Week Ending Feb 17th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Feb 20th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	908	7	0.8%	869	880	858	892	123.32	0.85	0.7%	120.06	120.58	117.93	122.93
IOPI58	58% Fe Fines	791	0	0.0%	813	798	761	793	108.10	-0.12	-0.1%	113.29	110.17	105.29	110.04
IOPI65	65% Fe Fines	1019	7	0.7%	981	992	970	1003	138.87	0.83	0.6%	136.05	136.43	133.84	138.73

IRON ORE SEABORNE INDEX (IOSI)

Feb 20th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	126.42	0.00	0.00%	107.27	122.31	83.90	159.45
IOSI65	65% Fe Fines	137.35	0.00	0.00%	117.64	139.45	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures open lower and go higher, the main contract i2305 closed 894, an increase of 1.02% throughout the day. Some traders were active to sell, some steel procurement more cautious, the continuation of a small number of multi-frequency procurement strategy. today's overall market transaction atmosphere in general.PBF at Shandong port dealt 895-900 yuan/mt; increased 5-10 yuan/mt from last Friday. PBF at Tangshan port dealt 905-910 yuan/mt; increased 8-10 yuan/mt from last Friday.

According to SMM, the global iron ore shipments totaled 27.02 million tons, up 7.8% week-on-week. Among them, Australia shipped 15.15 million tons, an increase of 7.1% over the previous week. Brazil shipped 5.87 million tons, up 5.8% week-on-week. However, affected by the reduction in the volume of Australia and Pakistan shipments in the early stage, the volume of arrivals in the current period was 28.52 million tons, down 1.06% week-on-week. Although the shipment volume has slightly increased this week, it remains at the annual low level. It is said in the market today that the heavy rain in Brazil has led to another flood in some regions, which may have an impact on the shipment of Brazilian

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Feb 17th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	990	-8	-0.8%	899	1036	770	1405	129.74	-1.99	-1.51%	115.69	138.89	94.72	199.13

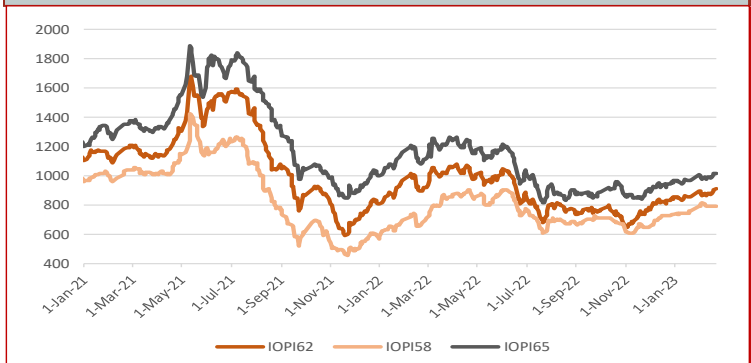
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Feb 17th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1036	-4.3%	779	1645	151.62	-4.34%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1055	1.0%	780	1630	154.40	0.96%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	790	0.0%	620	1310	115.62	0.00%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1059	-4.2%	800	1752	154.98	-4.25%	117.19	272.32

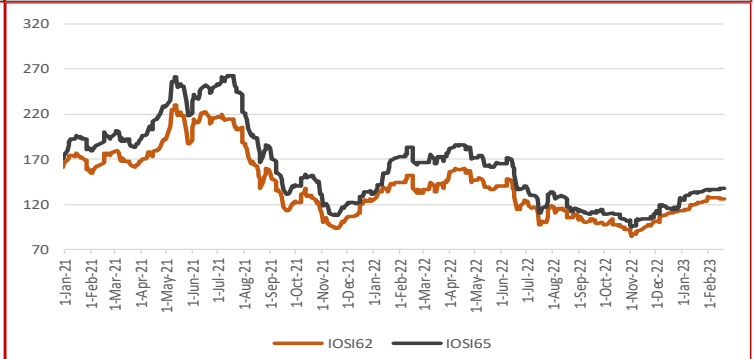
Week Ending Feb 17th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	937.76	-1.07%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.7849

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Feb 20th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	October	November	December	January	MTD	QTD	YTD	October	November	December	January	MTD	QTD	YTD
IOPI62	62% Fe Fines	736	714	822	856	869	881	880	95.56	92.18	109.43	117.00	120.06	120.90	120.58
IOPI58	58% Fe Fines	683	635	711	751	813	792	798	89.34	82.39	95.17	103.34	113.29	109.35	110.17
IOPI65	65% Fe Fines	918	865	935	968	981	993	992	120.08	112.37	124.90	132.80	136.05	136.72	136.43

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Feb 20th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	October	November	December	January	MTD	QTD	YTD
IOSI62	62% Fe Fines	95.80	93.05	108.92	119.76	107.27	107.62	122.31
IOSI65	65% Fe Fines	106.70	102.40	116.83	132.39	117.64	118.51	139.45

FREIGHT RATES

Feb 17th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	6.06	-0.01	-0.25%	3.57	16.77
Tubarao - Qingdao	C3	16.11	-0.06	-0.38%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Feb 17th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	October	November	December	January	MTD	QTD	YTD	October	November	December	January	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	877	811	922	964	899	900	1036	109.81	100.83	117.06	126.47	115.69	116.42	138.89

IRON ORE INDEX PREMIUMS/DISCOUNTS

Feb 20th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-117	-12.89%
IOPI65	65% Fe Fines	111	12.22%

Feb 20th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	10.93	8.65%

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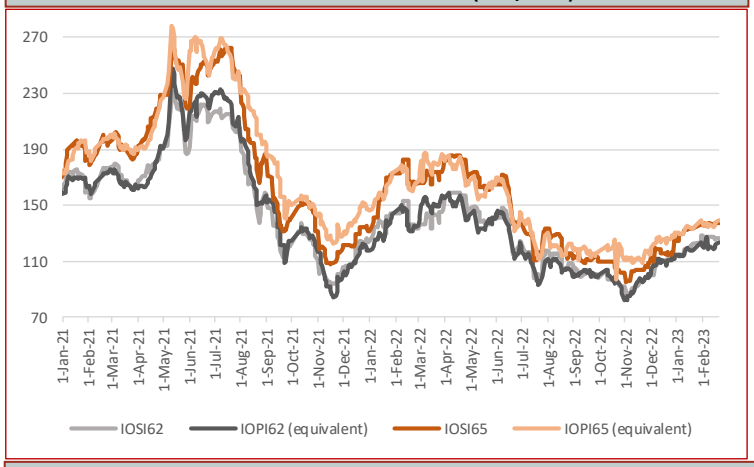
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Feb 20th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	851	7	-57
SIMEC Fines	781	7	-127
PB Fines	876	7	-32
Newman Fines	877	7	-31
MAC Fines	858	7	-50
Jimblebar Blended Fines	773	7	-135
Carajas Fines	1089	7	181
Brazilian SSF	873	8	-35
Brazilian Blend Fines	889	7	-19
RTX Fines	791	7	-117
West Pilbara Fines	819	7	-89

Feb 20th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	121.90	0.00	-4.52
SIMEC Fines	118.40	0.00	-8.02
PB Fines	122.70	0.00	-3.72
Newman Fines	125.55	0.00	-0.87
MAC Fines	122.70	0.00	-3.72
Jimblebar Blended Fines	115.05	0.00	-11.37
Carajas Fines	155.95	0.00	29.53
Brazilian SSF	130.15	0.00	3.73
Brazilian Blend Fines	131.05	0.00	4.63
RTX Fines	116.30	0.00	-10.12
West Pilbara Fines	125.65	0.00	-0.77

Feb 20th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	791	0	0
FMG Blended Fines	801	0	10
Robe River	802	1	11
Western Fines	804	0	13
Atlas Fines	798	0	7
Yandi	784	0	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	11.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	18.00	0.00
1% Alumina	High Fe Grade Al <2.25%	90.00	7.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	50.00	0.00
	Low Fe Grade Al 2.25-4%	20.00	-5.00
1% Silica	High Fe Grade Si <4%	5.00	0.00
	High Fe Grade Si 4-6.5%	22.00	0.00
	Low Fe Grade	15.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	5.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00
	Low Fe Grade 0.09<P<0.1%	14.00	6.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	2.00	0.25
	High Grade Fe 63 - 64%	4.00	0.00
	High Grade Fe 64 - 65%	4.00	0.00
	High Grade Fe 65 - 65.5%	4.00	0.00
1% Alumina	High Fe Grade Al <2.25%	2.50	1.00
	High Fe Grade Al 2.25-4%	2.00	1.00
1% Silica	High Fe Grade Si <4%	3.00	0.00
	High Fe Grade Si 4 - 6.5%	0.25	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	2.50	1.00
	High Fe Grade 0.115%<P<0.15%	4.50	0.25

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	5.00	0.00	Jingtang	-20.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

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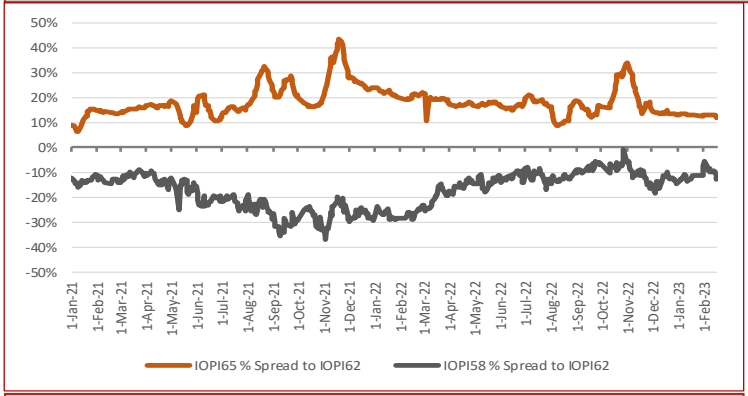
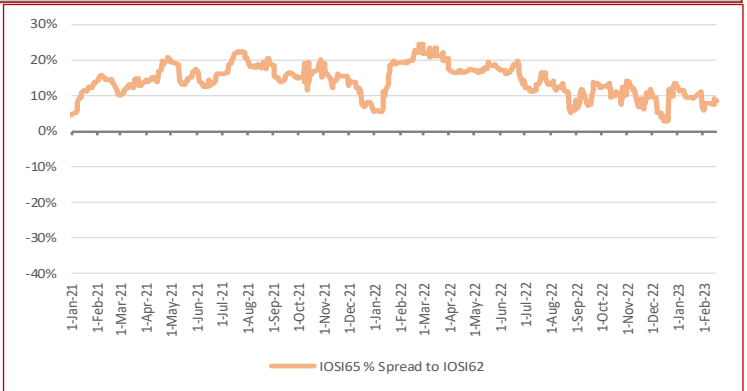
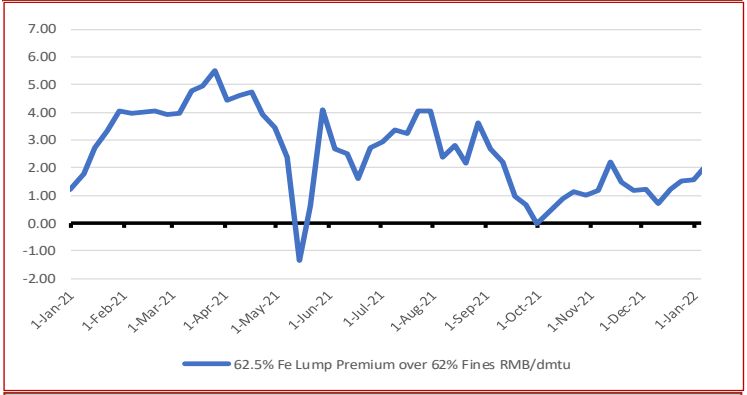
IRON ORE INDEX PREMIUMS/DISCOUNTS

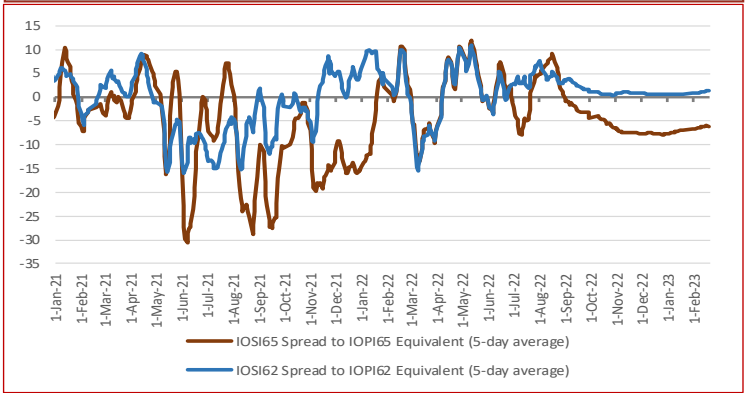




WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)

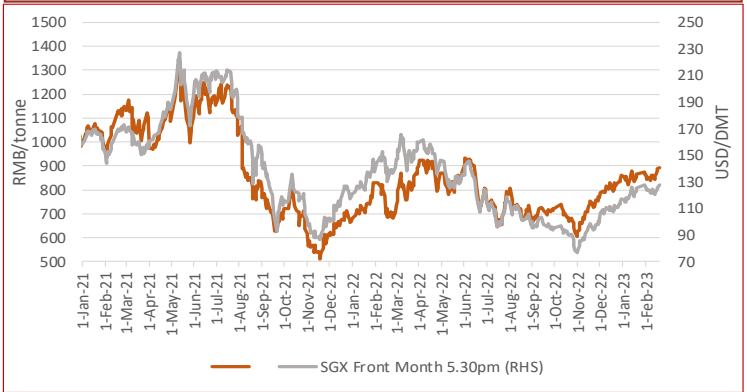


IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)

FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

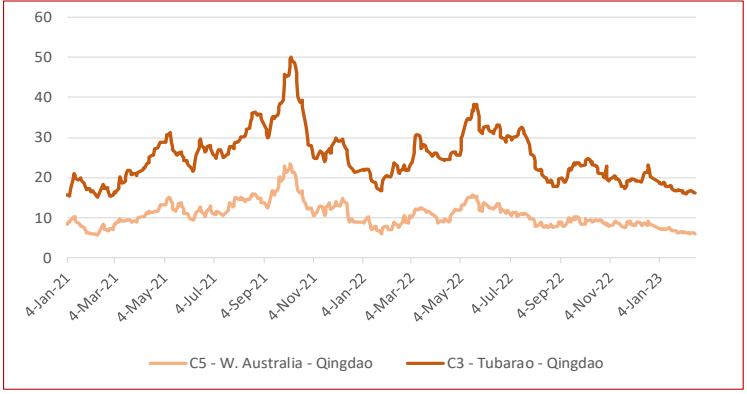
Week Ending Feb 17th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.14	1.34%	9.84	19.20
Qingdao	26.24	1.27%	9.41	26.24
Caofeidian	11.63	5.34%	9.05	16.29
Tianjin	11.88	0.25%	7.14	12.97
Rizhao	19.15	1.38%	9.44	19.26
Total (35 Ports)	137.11	0.75%	98.80	155.39

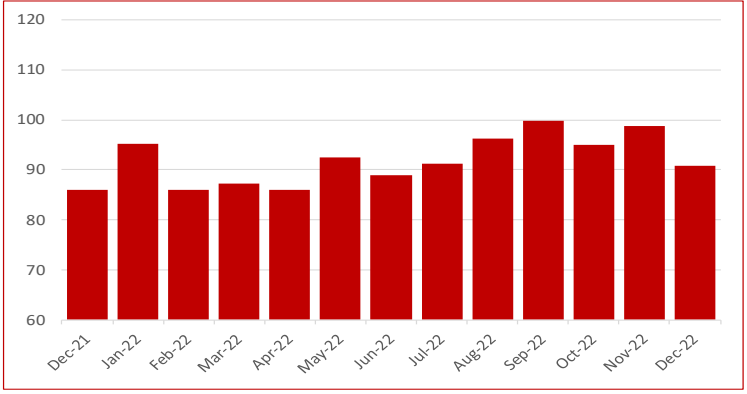
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Feb 20th, 3pm close			Feb 20th, 5:30pm		
Contract	I2305	Change	Change %	Mar. 23	Change	Change %
Closing Price	894.00	4.50	0.51%	127.75	2.45	1.96%
Vol traded ('000 lots)	59.99	-4.76	-7.35%	4.82	-0.17	-3.49%
Open positions ('000 lots)	94.58	0.92	0.99%	31.55	0.65	2.09%
Day Low	879.0	-0.50	-0.06%	128.75	4.30	3.46%
Day High	898.5	5.50	0.62%	125.20	-1.50	-1.18%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/2/17	Change	Change %
ReBar HRB400 ϕ18mm	4,180	40	0.97%
Wirerod Q300 ϕ6.5mm	4,430	40	0.91%
HRC Q235/SS400 5.5mm*1500*C	4,240	80	1.92%
CRC SPCC/ST12 1.0mm*1250*2500	4,620	30	0.65%
Medium & Heavy Plate Q235B 20mm	4,300	60	1.42%
GI ST02Z 1.0mm*1000*C	4,870	40	0.83%
Colour Coated Plate	7,580	80	1.07%

CHINESE STEEL MILL PROFITABILITY

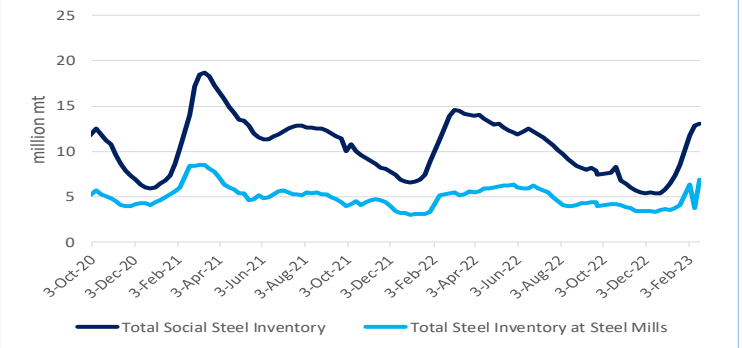
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	127.10	0	Mmi CFR Equivalent index for 1st Feb
Coke	2,960	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,150	-50	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,613	1	Q234, incl. tax
Rebar cost - Blast furnace	4,126	11	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	64	79	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,128	11	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	112	79	based on Shanghai prices, incl. tax

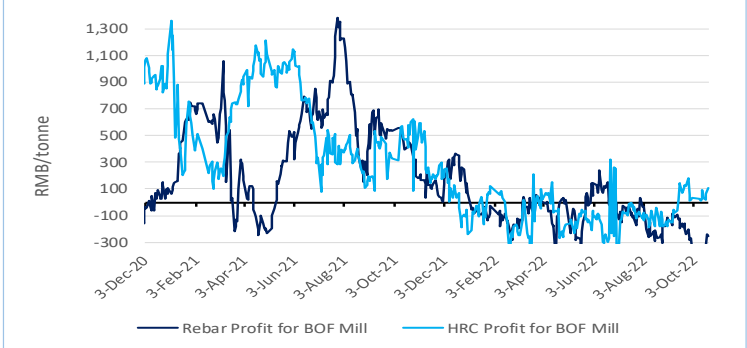
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

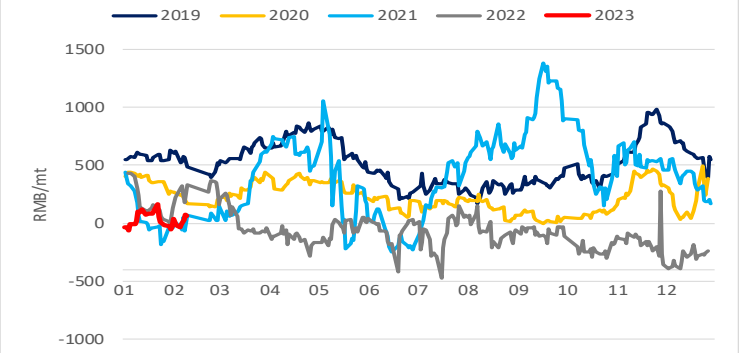
CHINESE STEEL INVENTORIES



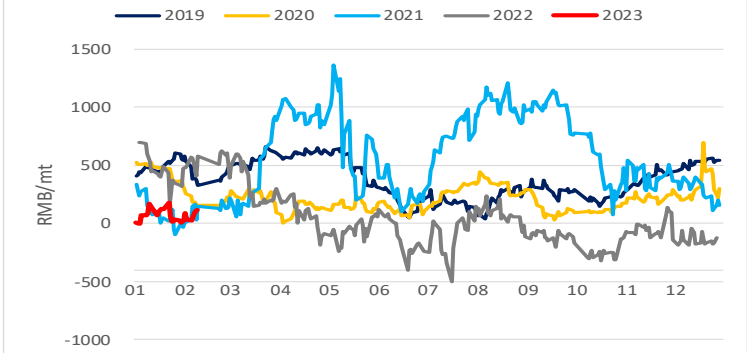
CHINESE STEEL MILL PROFITABILITY



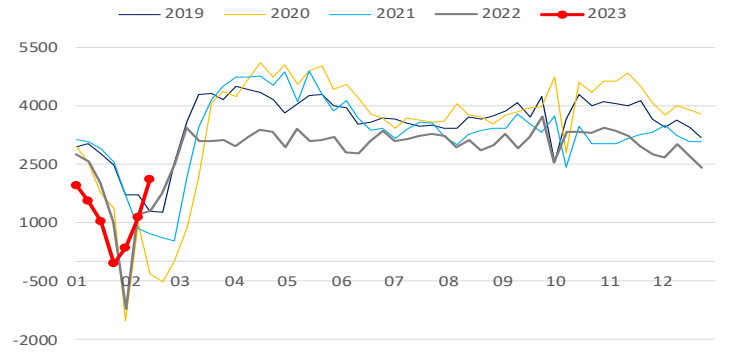
CHINESE STEEL MILL PROFITABILITY—Rebar



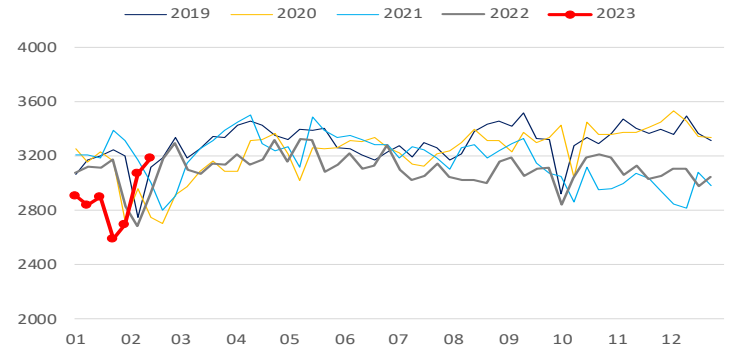
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



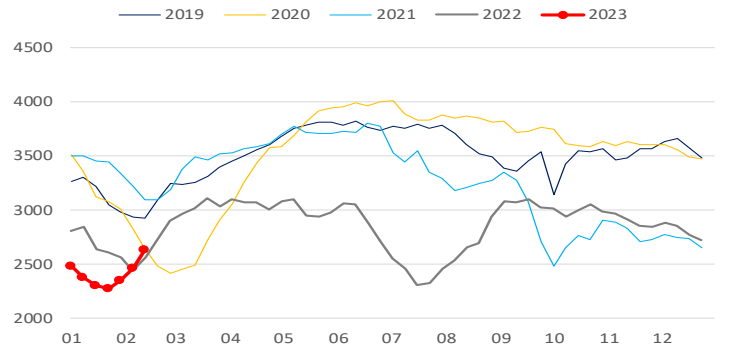
CHINESE STEEL CONSUMPTION—Rebar



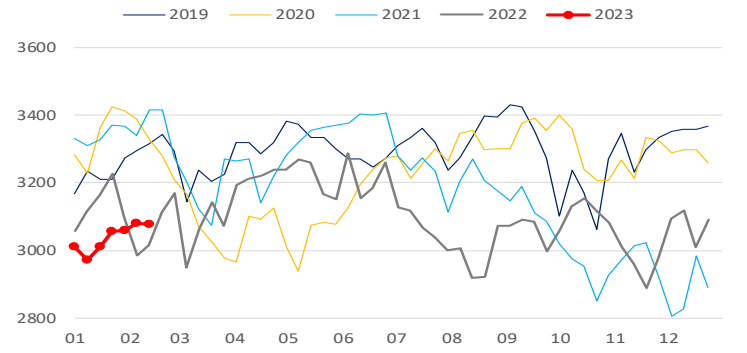
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Feb 20th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Feb 20th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

CONTACT US

MMI Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7629 E: prices@mmiprices.com
SMM Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7630 E: service.en@smm.cn
SMM Shanghai Office Details: 9th FL, Building 9, Lujiazui Software Park, No.20, Lane 91, Pudong, Shanghai Tel: +86 021 5155 0306 E: service.en@smm.cn

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